Machine Learning

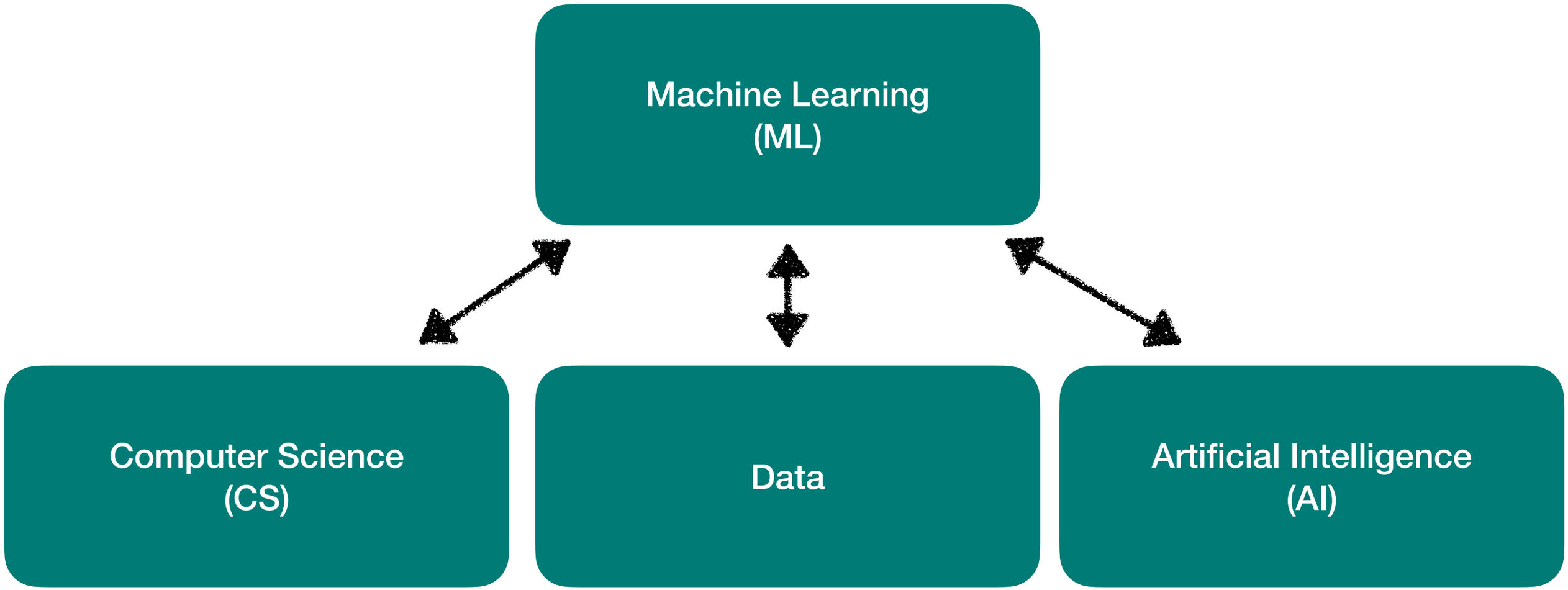
Introduction
Hoda Eldardiry

What is Machine Learning?

- **What is Machine Learning?**

- **What are some applications of Machine Learning?**

Connection & Comparison



CS & ML

- Traditional CS
  - Program the computer for every task

- “New” paradigm
  - Provide examples to the machine
  - Machine learns to accomplish a task based on examples

- ML*
  - Give computers the ability to learn without being explicitly programmed

* Definition by Arthur Samuel (1959)

Data & ML

- ML is the study of algorithms that:
  - Learn from large quantities of data
  - Identify patterns
  - Make predictions on new instances

AI & ML

- AI goal: develop intelligent machines
- **What defines intelligence?**

AI & ML

- AI goal: develop intelligent machines

- Subgoals to achieve human intelligence:
  - Perception
  - Reasoning
  - Control / Motion / Manipulation
  - Planning
  - Communication
  - Creativity
  - **Learning**

Intelligence/Learning
What can one learn?

What can one learn?

Learning to ..	AI/ML Field
Recognize spoken words	Speech Recognition
Drive an autonomous vehicle	Robotics
Play games	Games / Reasoning
Recognize images	Computer Vision

From data to knowledge

DATA

- **Data exists in many domains:**
  Computer Science
  Social Science
  Economics
  Medicine
  Bioinformatics

- **Example datasets:**
  WWW
  Social networks
  Biological networks
  Communication networks
  Transportation networks
  Energy grids
  Knowledge graphs
  Social media
  Finance data
  Citation data
  Marketing data
  Molecular structures

Learn from data

Complex data characteristics pose challenges in learning:

- Multimodal
- Multirelational
- Dynamic
- Large
- Noise

KNOWLEDGE

- Insights
- Patterns
- Inferences
- Predictions
- Correlations
- Dependencies
- Relations

ML big picture

- Learning paradigms - data form? prediction form?
- Problem formulation - structure of output prediction?
- Theoretical foundations - what principles guide learning?
- Facets of building ML systems - how to build systems?
- Big ideas in ML - ideas driving development in ML?
- Applications of ML - key challenges?

Learning paradigms

- **What data is available and when? What form of prediction?**

Technique	Brief description
Supervised learning	Learn from data and labeled examples
Unsupervised learning	Learn from data
Semisupervised learning	Learn from labeled examples and data
Reinforcement learning	Learn from trial and error
Active learning	Learn which examples to learn from
Ensemble learning	Learn from multiple models
Online learning	Learn in real-time - dynamically adapt to new data
Recommender systems	Predict user rating or preference of items - learn from user info, item info, similar users
Feature learning	Learn features from data - features are then used to learn from data
Hyperparameter optimization	Tuning - choosing optimal parameters that control the learning process

Problem formulation

Structure of output prediction	Problem formulation
Boolean	Binary Classification
Categorical	Multiclass Classification
Ordinal	Ordinal Classification
Real	Regression
Ordering	Ranking
Multiple discrete	Structured Prediction
Multiple continuous	Example: dynamical systems
Both discrete & continuous	Example: mixed graphical models

Theoretical foundations

- What principles guide learning?
    - Probabilistic
    - Information theoretic
    - Evolutionary search
    - ML as optimization

Facets of building ML systems

- How to build systems that are ***robust, efficient, adaptive, effective***?
  - Data preparation
  - Model selection
  - Training (optimization / search)
  - Hyperparameter tuning on validation data
  - (Blind) Assessment on test data

Big ideas in ML

- What are the **ideas** driving **development** of the ML field?
  - Inductive bias
  - Generalization / overfitting
  - Bias-variance decomposition
  - Generative vs. discriminative
  - Deep nets, graphical models
  - PAC learning
  - Distant rewards

Applications of ML

- Key challenges..

  - **Natural Language Processing (NLP)**
    – Machine translation, question answering, dialog systems

  - **Speech recognition**
    – Siri, Cortana

  - **Computer Vision**
    – Image and video analysis

  - **Robotic Control**
    – Autonomous vehicles

  - **Intelligent assistants**
    – Activity recognition, recommender systems

  - **Computational finance**
    — Stock trading, portfolio optimization

DEFINING LEARNING PROBLEMS

Learning problems

- Learning components
  —Task, T
  —Performance measure, P
  —Experience, E

- Learning definition*
  A computer program **learns**
  if its **performance**, measured by **P**, at **tasks in T**,
  improves with **experience E**



*Tom Mitchell
1997

Learning example

Identify the learning components..

[choices: task, experience, performance measure]

Problem: Design a spam filter

- Classify emails as spam or not spam […]
- Watch you label emails as spam or not spam […]
- Number of emails correctly classified as spam/not spam […]

Learning example

Problem: Design a spam filter

- Classify emails as spam or not spam
  Task

- Watch you label emails as spam or not spam
  Experience

- Number of emails correctly classified as spam/not spam
  Performance measure

CAPTURING EXPERT KNOWLEDGE

Capturing expert knowledge

Problem — Learning to respond to voice commands (Siri)

Solution #1 — Expert systems

- **20 years ago:** rule-based system
- Ask the expert to:
  - Obtain a PhD in Linguistics
  - Introspect about language structure
  - Write down the rules they devise

Voice command	Rule to respond to voice command
Give me directions to Starbucks	If: “give me directions to X” Then: directions(here, nearest(X))
How do I get to Starbucks?	If: “how do i get to X” Then: directions(here, nearest(X))
Where is the nearest Starbucks?	If: “where is the nearest X” Then: directions(here, nearest(X))
I need directions to Starbucks	If: “I need directions to X” Then: directions(here, nearest(X))
Is there a Starbucks nearby?	If: “Is there an X nearby” Then: directions(here, nearest(X))
Starbucks directions	If: “X directions” Then: directions(here, nearest(X))

Capturing expert knowledge

Problem — Learning to respond to voice commands (Siri)

- **Experts**
  — Very good at answering questions about specific cases
  — Not very good at telling **how** they do it

- **1990s**
  — Why not just have experts tell you what they do on **specific cases**
  — Then let **ML** tell you how to come to the same decisions that they did

- **Solution #2 — Annotate Data and Learn**
  — Collect raw sentences $\{x_1, \ldots, x_n\}$
  — Experts annotate their meaning $\{y_1, \ldots, y_n\}$

Raw sentence x_i	Annotation y_i
x_1: How do I get to Starbucks?	y_1: directions(here, nearest(Starbucks))
x_2: Show me the closest Starbucks	y_2: map(nearest(Starbucks))
x_3: Send a text to John that I'll be late	y_3: txtmsg(John, I'll be late)
x_4: Set an alarm for seven in the morning	y_5: setalarm(7:00AM)

Learning problems

- **Identify T, P and E for the two learning problems below**

- **Learning to play chess**
  Task, T:
  Performance measure, P:
  Experience, E:

- **Learning to respond to voice commands (example: Siri)**
  Task, T:
  Performance measure, P:
  Experience, E:

Learning problems

- Learning to play chess
  Task, T: playing chess
  Performance measure, P: probability program wins next game
  Experience, E: experience of playing many games

- Learning to respond to voice commands (Siri)
  Task, T: predict action from speech
  Performance measure, P: % correct actions taken
  Experience, E: examples of (speech, action) pairs

Learning tasks

- Select a **task** from the list below
  Define: **performance measure** and **experience**

Example Tasks

—Identify objects in an image
—Translate from one human language to another
—Assess risk (e.g. in loan application)
—Make decisions (e.g. in loan application)
—Assess potential (e.g. in admission decisions)
—Categorize a complex situation (e.g. medical diagnosis)
—Predict outcome (e.g. medical prognosis, stock prices, inflation, temperature)
—Predict events (default on loans, quitting school, war)
—Plan ahead under partial knowledge (Poker, Bridge)
—Email spam filtering
—Plan ahead under perfect knowledge (chess)
—Recognize speech

ML Problem formulation

- The same **problem** can be formulated in various ways (ML Tasks)

- Example **learning problem**: Process loan applications
  – Task: Compute credit score (regression)
  – Task: Compute probability of default (density estimation)
  – Task: Generate loan decision (classification)

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Project Learning Components

Think about a project topic

- **What are the learning problem(s) involved?**

- **For each learning problem, list the learning components**
  - **Task?**
  - **Performance measure?**
  - **Experience?**

Reminders

- Join a Project Team